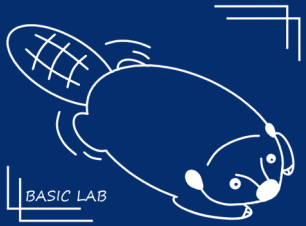


● MoBiQuant: problem setting and research gap

• 1. Static PTQ

- GPTQ / OmniQuant / SmoothQuant / AWQ optimize one chosen bit-width.
- Strong at a fixed precision, but brittle when runtime precision changes.



BASIC LAB



- Core observation: outlier migration breaks cross-bit generalization
- Tokens causing high quantization error are not stable across bit-widths.



● Method: many-in-one bit slices plus token-aware routing

• 1. MoBiSlice

- Recursively quantize residual weights into fixed 2-bit slices.
- Reconstruct 2/4/6/8-bit weights by summing slices.



● Main results and conclusion

• 1. Accuracy

- Matches or surpasses static scalar PTQ on LLaMA2/LLaMA3 families.
 - - Large gains at 2-3 bit when compared with any-precision baselines.

